

oda class

MATHEMATICS

— STUDY MATERIAL —



CLASS 6



Compendium

A. Introduction

I. Making numbers from digits

B. Comparing numbers

I. Use of commas

C. Quick estimation of numbers

I. Quick estimation of Sum and Difference

II. Quick estimation of Product & Quotient

D. Using brackets (BODMAS)

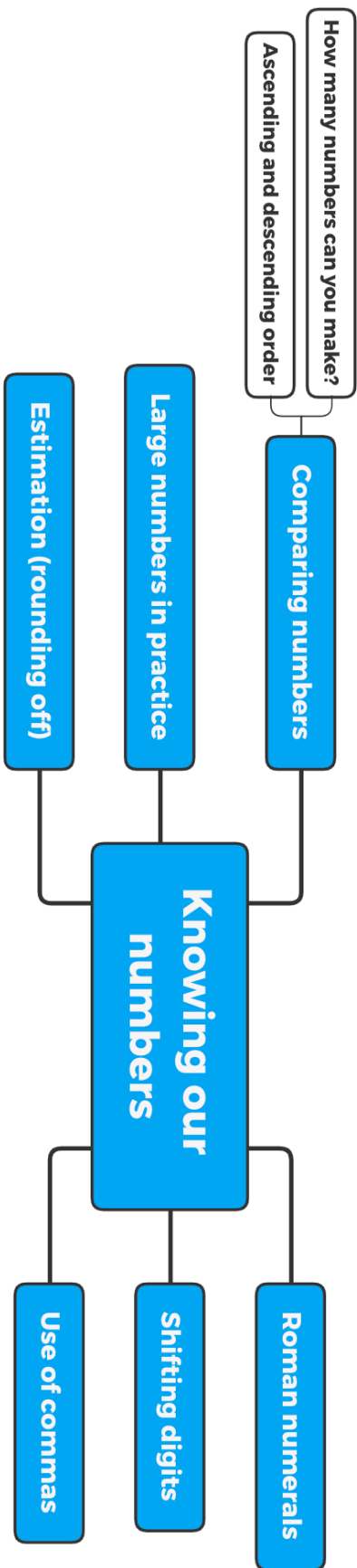
E. Roman numerals

F. Bird's-eye view

G. Solved examples

H. Practice yourself

I. Solutions



CHAPTER 1

KNOWING OUR NUMBERS

A. Introduction

Have you ever thought about how a number is formed?

Knowing our numbers helps us in counting objects in large numbers & representing them through numerals. It helps in communicating through suitable number names & to count concrete objects.

NOTATION



Writing a number in figures

NUMERATION



Writing a number in words.

Numbers are formed with the help of digits.

1 2 3 4 5
6 7 8 9 0

Making numbers from digits

To form different numbers from given digits,

- No digit must be repeated
- All the given digits must be used

Example 1

46 is formed with digit 4 and 6

Whereas 64 is also formed with the same digits

Example 2

Take four digits 2, 8, 4, 5

The greatest number we get is 8542

The smallest number we can get is 2458



Checkpoint

1. Form greatest number with 6,4,0,9
2. Form smallest number with 8,4,2,3
3. Form greatest number with 3,0,9,7,5
4. Find difference between the greatest and smallest number formed with digits 3, 5, 9,6
5. Find difference between the greatest and smallest number formed with digits 2,0,7,4,8.

So in this chapter, we will talk about numbers and operations on it.



Mental floss!

The concept of **zero** as a digit was discovered in India.



CHAPTER 1

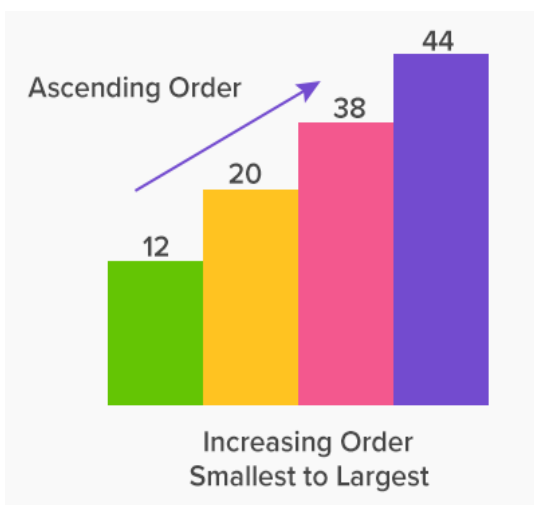
KNOWING OUR NUMBERS

B. Comparison of Numbers

The arrangements of numbers from the smallest to the greatest is called **ascending order**.

Example 3: 16, 43, 88, 97 are in ascending order.

Example 4: 743, 872, 1056, 6458 are in ascending order



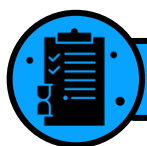
The arrangement of numbers from the greatest to the smallest is called **descending order**.

Example 5: 32, 21, 17, 13 are in descending order.

Example 6: 4232, 3843, 2841, 1513, 817 are in descending order



From left to right the height of kids are in ascending order i.e. shortest to tallest.



Checkpoint

1. Arrange in ascending order 6,4,0,9
2. Arrange in descending order 8,4,2,3
3. Arrange in ascending order 7783, 8232, 5124, 9212.
4. Arrange in descending order 8325, 92737, 6532, 32535, 523443

Use of commas

If two numbers have an unequal number of digits, then the number with the **greater number of digits is greater**.

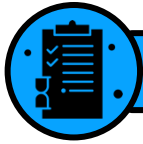
Commas added to numbers help us read and write large numbers easily. As per **Indian Numeration**, the first comma is placed after hundreds place; then commas are placed after every two digits.

INDIAN NUMBER SYSTEM Initial 3-digit split followed by 2-digit split thereafter			
$10^3 = \text{Thousand}$ $1,000$ 3	$10^5 = \text{Lakhs}$ $1,00,000$ 2 3	$10^7 = \text{Crore}$ $1,00,00,000$ 2 2 3	$10^9 = \text{Arab}$ $1,00,00,00,000$ 2 2 2 3

Example: 12, 34, 567 and many more.

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Checkpoint

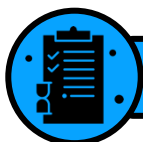
1. Write the following in figures in the Indian system:

- (i) eight thousand : 8,000
- (ii) Sixty thousand : 60,000
- (iii) Seven lakh : 7,00,000
- (iv) Thirty lakh : 30,00,000
- (v) Nine crore : 9,00,00,000

As per **International Numeration**, the first comma is placed after hundreds place; then commas are placed after every three digits.

INTERNATIONAL NUMBER SYSTEM 3-digit split in the numbers

$10^3 = \text{Thousand}$	$10^6 = \text{Million}$	$10^9 = \text{Billion}$	$10^{12} = \text{Trillion}$
1,000 3	1,000,000 3 3	1,000,000,000 3 3 3	1,000,000,000,000 3 3 3 3



Checkpoint

1. Arrange the numbers below in ascending order :

- (a) 847, 9654, 8320, 571 (b) 9851, 25751, 36501, 38802

2. Arrange the numbers below in descending order :

- (a) 5000, 7500, 85400, 7861 (b) 1971, 45321, 88715, 92547

3. Find the greatest and the smallest numbers.
 - 4536, 4892, 4370, 4452
 - 15623, 15073, 15189, 15800
 - 25286, 25245, 25270, 25210
 - 6895, 23787, 24569, 24659
4. Use the given digits without repetition and make the greatest and smallest 4-digit numbers.
 - (a) 2, 8, 7, 4
 - (b) 9, 7, 4, 1
 - (c) 4, 7, 5, 0
 - (d) 1, 7, 6, 2
 - (e) 5, 4, 0, 3
5. Insert commas wherever required and write the names according to Indian System of Numeration :
 - (a) 87595762
 - (b) 8546283
 - (c) 99900046
 - (d) 98432701
6. Insert commas wherever required and write the names according to International System of Numeration :
 - (a) 78921092
 - (b) 7452283
 - (c) 99985102
 - (d) 48049831

C. Quick estimation of numbers

A reasonable approximation of the actual value is called an **estimate**. A rough, however, a quick estimate of the result of number operations can be performed by rounding off the numbers is involved.



Try and learn

Estimating numbers to the nearest tens is performed by rounding off numbers 1, 2, 3 and 4 to 0 and number 6, 7, 8, 9 to 10.

(i) 79

Solution:

Unit or ones digit in 79 is 9, which is greater than 5.

So, we replace the ones digit by 0 and increase the tens digit by 1 to get the rounded-off number.

Hence, rounded off number = 80.

(ii) 44

Solution:

The given number is 44.

Its ones or unit digit is 4, which is less than 5. So, we replace the ones digit by 0 to get the rounded-off number.

Hence, rounded off number = 40.

Estimating numbers to the closest hundreds is done by rounding off numbers 1 to 49 to 0 and numbers 51 to 99 to 100.

(iii) 5839

Solution:

The given number is 5839.

Its tens digit is 3, which is less than 5. So, we replace each of the tens and one's digits by 0 and keep the other digits as they are to round off the given number to nearest hundreds.

Hence, 5839 is rounded off to nearest hundreds as 5800.

(ii) 9472

Solution:

The given number is 9472.

Its tens digit is 7, which is greater than 5. So, we increase the digits at hundreds places by 1 and replace each one of the digits at tens and one's place by 0 to round off the given number to nearest hundreds.

Hence, 9472 is rounded off to nearest hundreds as 9500.

Estimating numbers to the closest thousands is done by rounding off numbers 1 to 499 to 0 and the numbers 501 to 999 to 1000.

(i) 14329

Solution:

The given number is 14329.

Its digit at hundreds place is 3, which is less than 5. So, first, we replace each of the hundreds, tens and one's digits by 0 and then keep the other digits as they are.

So, the number 14329 is rounded off to nearest thousands as 14000.

(ii) 14729

Solution:

The given number is 14729.

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Its digit at hundreds place is 7, which is greater than 5. So, we increase the digit at thousands place by 1 and replace each one of the digits at hundreds, tens and ones place by 0 to get the rounded-off numbers.

Hence, the number 14729 is rounded off to nearest thousands as 15000.

Quick estimation of sum and difference

The procedure of estimating either sum and difference are based on doing estimation of each number to nearest hundred, thousand or as asked and then we perform the operation of sum or difference . let's see following examples.



Try and learn

Example 7:

Estimate the sum of 5290 and 17986 by estimating the numbers to their nearest

(i) Hundreds

(ii) Thousands

Solution:

(i) Estimating the numbers 5290 and 17986 to their nearest hundreds, we get 5300 and 18000 respectively.

We have, $5300 + 18000 = 23300$

Therefore, estimated sum is 23300.

Example 8:

Estimate: $5673 - 436$ by rounding off the numbers to their greatest places. Also, find a reasonable estimate.

Solution:

We have, $5673 - 436 = 5237$.

The greatest place in 5673 is thousands place, and in 436 the greatest place is hundred place.

Estimating 5673 to nearest thousands, we get 6000

Estimating 436 to nearest hundreds, we get 400

Therefore estimated difference = $6000 - 400 = 5600$

Clearly, it is not closer to the actual difference. So, it is not a reasonable estimate.

Let us round off 5673 and 436 to nearest hundreds.

5673 rounds off as 5700.

436 rounds off as 400

Therefore estimated difference = $5700 - 400 = 5300$

Quick estimation of product & quotient

The procedure of estimating either product and quotient are based on doing estimation of each number to nearest hundred, thousand or as asked and then we perform the operation of product and quotient. Let's understand it through the following examples.

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Try and learn

Example 9: Estimate the product 958×387 by rounding off each factor to its greatest place.

Solution:

By simple observation, we can infer that each factor is a three-digit number. So, we round off each factor to nearest hundreds.

958 rounds off as 1000

387 rounds off as 400

Therefore estimated product = $1000 \times 400 = 400000$

Example 10: Find the estimated quotient $2838 \div 125$ by rounding off the numerator and denominator to the nearest hundreds.

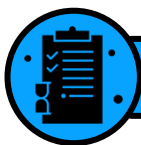
Solution:

We find that

2838 rounds off to nearest hundreds 2800

125 rounds off to nearest hundreds as 100

Therefore, estimated quotient = $2800 \div 100 = 28$.



Checkpoint

1. Estimate: $5,290 + 17,986$.
2. Estimate: $5,673 - 436$.
3. Estimate the following products :

(a) 87×313	(b) 9×795
(c) 898×785	(d) 958×387
4. Estimate each of the following calculations using general rule:

(a) $730 + 998$	(b) $796 - 314$
(c) $12,904 + 2,888$	(d) $28,292 - 21,496$
5. Estimate the following products using general rule:

(a) 578×161	(b) 5281×3491
(c) 1291×592	(d) 9250×29
6. Estimate $74347 \div 7236$ to nearest thousand.
7. Estimate $929371 \div 9437$ to nearest thousand.

D. Importance & usage of brackets (BODMAS)

Brackets are tools to simplify any expansion containing more than one mathematical operation. In an expression that contains brackets, the values inside the brackets must be simplified into one single number first, then the operations outside the brackets should follow.

1. We perform various functions on the numbers such as addition, subtraction, multiplication, division etc.

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2. Sometimes we may have to execute many functions together without knowing their correct sequence. To avoid errors while doing so, we use brackets.
3. The numbers on which we perform the same operation are packed into brackets.
4. While solving the problems with brackets, first simplify the term inside the brackets.

BODMAS stands for **B**racket, **O**f, **D**ivision, **M**ultiplication, **A**ddition, and **S**ubtraction. The **BODMAS** is used to explain the order of operation of a mathematical expression.



Try and learn

Example 11: $37 - [5 + \{28 - (19 - 7)\}]$

Solution: $37 - [5 + \{28 - (19 - 7)\}]$

$= 37 - [5 + \{28 - 12\}]$ [Removing the inner most bracket ()]

$= 37 - [5 + 16]$ [Removing braces { }]

$= 37 - 21$

$= 16$

Example 12: $78 - [24 - \{15 - (5 - 3 - 1)\}]$

Solution: $78 - [24 - \{15 - (5 - 4)\}]$

$= 78 - [24 - \{15 - 1\}]$ [Removing parentheses]

$= 78 - [24 - 14]$ [Removing braces]

$= 78 - 10$

$= 68.$



Checkpoint

1. $94 - [32 - \{25 - (6 - 4 - 1)\}]$

2. $118 - [14 - \{25 - (8 - 3 - 2)\}]$

E. Roman numerals

In Roman numerals, a symbol is never repeated more than three times, but the symbols **V**, **L** and **D** are never repeated.

Rules of representation of Roman Numerals:

Roman numerals should be read from left to right, and the letters of Roman numerals are arranged from the largest to the smallest.

If any symbol representing smaller value is written to the right of a symbol representing greater value, then its value gets added to the value of the greater symbol. If a symbol representing smaller value is written to the left of a symbol of greater value, its then value is subtracted from the value of the greater symbol.

The symbol —I can be subtracted from —V and —X only.

The symbol —X can be subtracted from —L, —M and —C only.

In Roman numerals

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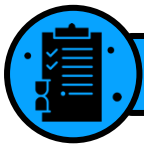
30 is written as XXX,

50 is written as L

80 is written as LXXX

90 is written as XC

100 is written as C



Checkpoint

1. XLVII =

2. CXVI =

3. MCXX =

4. MCMXIV=

5. 1078 =

6. 840 =



Mental floss!

Another name of zero is a cypher.



Bird's-eye view

1. To form different numbers from given digits,
 - No digit must be repeated
 - All the given digits must be used
2. The arrangements of numbers from the smallest to the greatest is called ascending order.

3. The arrangement of numbers from the greatest to the smallest is called descending order.
4. If two numbers have an unequal number of digits, then the number with the greater number of digits is greater.
5. As per International Numeration, the first comma is placed after hundreds place; then commas are placed after every three digits.
6. A reasonable approximation of the actual value is called an estimate
7. The procedure of estimating either sum and difference are based on doing estimation of each number to nearest hundred, thousand or as asked and then we perform the operation of sum or difference
8. The procedure of estimating either product and quotient are based on doing estimation of each number to nearest hundred, thousand or as asked and then we perform the operation of product and quotient.
9. BODMAS stands for Bracket, Of, Division, Multiplication, Addition, and Subtraction. The BODMAS is used to explain the order of operation of a mathematical expression.
10. In Roman numerals, a symbol is never repeated more than three times, but the symbols V, L and D are never repeated.
11. The symbol —I can be subtracted from —V and —X only.
12. The symbol —X can be subtracted from —L, —M and —C only.



Solved examples

Example 13:

The product of the place values of two 2's in 428721 is

- (a) 4 (b) 40000
(c) 400000 (d) 40000000

Solution : (c) The given number is 428721.

According to the question,

The place value of first 2 = $2 \times 10 = 20$

[it is at ten's place]

and the place value of second 2 = $2 \times 10000 = 20000$

[it is at ten thousand's place]

Their product = $20 \times 20000 = 400000$

Hence, the product of the place values of two 2's in 428721 is 400000.

Example 14:

$3 \times 10000 + 7 \times 1000 + 9 \times 100 + 0 \times 10 + 4$ is the same as

- (a) 3794 (b) 37940
(c) 37904 (d) 379409

Solution:

(c) We have,

$$\begin{aligned} & 3 \times 10000 + 7 \times 1000 + 9 \times 100 + 0 \times 10 + 4 \quad \text{[by simplification]} \\ & = 30000 + 7000 + 900 + 0 + 4 = 37904 \end{aligned}$$

Example 15:

If 1 is added to the greatest 7-digit number, it will be equal to

(a) 10 thousand (b) 1 lakh (c) 10 lakh (d) 1 crore

Solution:

(d) Greatest 7-digit number = 9999999

On adding 1 to greatest 7-digit number, we get

$9999999 + 1 = 10000000$, = 1 crore

Example 16:

The expanded form of the number 9578 is

(a) $9 \times 10000 + 5 \times 1000 + 7 \times 10 + 8 \times 1$

(b) $9 \times 1000 + 5 \times 100 + 7 \times 10 + 8 \times 1$

(c) $9 \times 1000 + 57 \times 10 + 8 \times 1$

(d) $9 \times 100 + 5 \times 100 + 7 \times 10 + 8 \times 1$

Solution:

(b) Given number is 9578.

On expanding 9578, we get $9578 = 9 \times 1000 + 5 \times 100 + 7 \times 10 + 8 \times 1$

[writing each digit as its place value]

Example 17:

5 When rounded off to nearest thousands, the number 85642 is

(a) 85600 (b) 85700

(c) 85000 (d) 86000

Solution:

(d) 85642 rounded off to nearest thousands = 86000

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Example 18:

The successor of 1 million is

- (a) 2 million (b) 1000001
(c) 100001 (d) 10001

Solution:

(b) To get successor of a number, we add 1 to the given number.
So, the successor of 1 million = $1000000 + 1 = 1000001$

Example 19:

Number of even numbers between 58 and 80 is

- (a) 10 (b) 11
(c) 12 (d) 13

Solution:

(a) Since, even numbers between 58 and 80 are 60, 62, 64, 66, 68, 70, 72, 74, 76 and 78.

Total even numbers between 58 and 80 = 10

Example 20:

Sum of the numbers of primes between 16 to 80 and 90 to 100 is

- (a) 20 (b) 18
(c) 17 (d) 16

Solution:

(c) Since, prime numbers between 16 to 80 are 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73 and 79.

Total prime numbers between 16 to 80 = 16 and prime numbers
between 90 to 100 = 97

Total number of prime numbers between 90 to 100 = 1

Sum of the numbers of primes

between 16 to 80 and 90 to 100 = $16+1 = 17$

Example 21:

How many crores makes a billion?

Solution:

1 billion = 100 crore

Example 22:

Find the greatest number from below number 763298 and 764298.

Solution :

764298 is the greatest number

Example 23:

Place commas correctly and write the numerals :

(a) Seventy three lakh seventy five thousand three hundred seven.

(b) Nine crore five lakh forty one.

(c) Seven crore fifty two lakh twenty one thousand three hundred two.

(d) Fifty eight millions four hundred twenty three thousand two hundred two.

Solution:

(a) 73,75,307

(b) 9,05,00,041

(c) 7,52,21,302

(d) 58,423,202

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Example 24:

On Sunday 6000 people visited a zoo. Among them 2615 were children then how many adult visited zoo?

Solution:

Total number of people visited the zoo = 6000

people Total number of children = 2615

children Then number of adults = $6000 - 2615 = 3385$

Example 25:

Kirti Bookstore sold books worth Rs.2,85,891 in the first week of June and books worth Rs.4,00,768 in the second week of the month. How much was the sale for the two weeks together? In which week was the sale greater and by how much?

Solution:

Sale of books in the first week = Rs.2,85,891.

Sale of books in the second week = Rs.4,00,768

Sale for the two weeks together = $\text{Rs.}2,85,891 + \text{Rs.}4,00,768 = \text{Rs.}6,86,659$.

The sale was greater in the second week by $\text{Rs.}4,00,768 - \text{Rs.}2,85,891$ i.e., by Rs.1,14,877.

Example 26:

Find the greatest and the smallest number. 42375, 42367, 42329, 42338.

Solution:

The greatest number is 42375.

The smallest number is 42329

Example 27:

Write and solve the expression: Thirteen multiplied by sum of four and eleven. Now reverse the result and add it to earlier result, what you obtain multiply it by 13.

Solution:

Expression $13 \times (4 + 11) = 13 \times 15 = 195$

On reversing the answer = 591

Adding reversed number to the result $591 + 195 = 786$

Multiplying it by 13 i.e. $786 \times 13 = 10218$.



NOVICE LEVEL

Very short answer type (Each carries one mark)

1. Arrange the following numbers in ascending order.
7561 1796 2356 3794 3010
2. Arrange the following numbers in ascending order:
6392, 6782, 6789, 6654
3. Arrange the following numbers in descending order.
9430 1070 1924 2010 2019
4. Estimate: $9,431 - 813$.
5. Estimate each of the following values by rounding off each number to closest hundreds:
(a) $874 + 478$ (b) $793 + 397$
(c) $11244 + 3507$ (d) $17677 + 13589$
6. What must be subtracted from 11010101 to get 2635967.
(a) 934134 (b) 7383414
(c) 8374134 (d) 937414
7. The difference between the place value and face value of 4 in 2416 is.
(a) 404 (b) 396
(c) 3000 (d) 2996
8. In the international system of numeration we write one billion for
(a) 1 crore (b) 10 crore
(c) 100 crore (d) 1000 crore

Short answer type (Each carries two / three marks)

9. How many different 3-digit numbers can be formed by using the digits 0, 2, 5 without repeating any digit in the number?
10. Find all three-digit numbers, which can be formed by using the digits 5, 3, 0 without repeating any digits.
11. Find the difference between the greatest and the least 5-digit number that can be written using the digits 6, 2, 7, 4, 3 each only once.
12. Write 645340001 using comma at required spaces in the International System of Numeration.
13. Which digits have the same face value and place value in 92078634.
14. Place value of 8 in 9,83,04,600 is _____.
15. How can the number 3,53,552 be read out according to Indian System?
16. What is the difference between the two place values of "1" in 3116365?
17. Write the following expression using the proper bracket: Multiply the difference of eight and two by five.
18. Simplify: $197 - [1/9 \{42 + (56 - 8 + 9)\} + 108]$
19. Simplify : $15 - (-5) \{4 - 7 - 3\} \div [3\{5 + (-3) \times (-6)\}]$
20. Simplify : $95 - [144 \div (12 \times 12) - (-4) - \{3 - 17 - 10\}]$

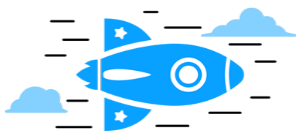
Long answer type (Each carries four marks)

21. If the distance between the home and a playground is 2 km 375 m, find the total distance covered by him in five days, if he walks both ways.

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22. A and B worked as a salesperson at a book store. They sold 6283 storybooks in all. A sold 3324 storybooks. How many storybooks were sold by B?
23. The population of a city was 9,75,688. In the first year, it increased by 4562, and in the second year, it decreased by 8876. What was the population of the town at the end of the second year?
24. If a table costs Rs. 350 and a chair costs Rs. 175, find the total amount needed to buy 30 table and 50 chairs. ?
25. A Game zone was held for four consecutive days in a school. The number of tickets sold at the ticket counter on the first, second, third and fourth day was respectively 1091, 1802, 1050 and 1751. Find the total number of tickets sold during those four days?



ADVANCE LEVEL

Fill in the blank

- (a) 10 million = _____ crore.
- (b) 10 lakh = _____ million.
- (c) 1 metre = _____ millimetres.
- (d) 1 centimetre = _____ millimetres. (e) 1 kilometre = _____ millimetres.
- (f) 1 gram = _____ milligrams.
- (g) 1 litre = _____ millilitres.
- (h) 1 kilogram = _____ milli grams.
- (i) 100 thousands = _____ lakh.

- (j) Height of a person is 1m 65cm. His height in millimetres is_____.
- (k) Length of river 'Narmada' is about 1290km. Its length in metres is_____.
- (l) The distance between Srinagar and Leh is 422km. The same distance in metres is_____.
- (m) Arranging of numbers from the largest to the smallest is called an arrangement in _____ order.
- (n) By reversing the position or order of digits of the greatest number made by five different non-zero digits, the new number is the _____ number of five digits.
- (o) By adding 1 to the largest_____ digit number, we will get ten lakh.
- (p) The number five crore twenty four lakh seventy eight thousand four hundred one can be written, using commas, in the Indian System of Numeration as _____.
- (q) In Roman Number System, the symbol X can be subtracted from_____, M and C only.
- (r) The number 66 in Roman numerals is_____.(s) The population of Pune was 2,538,473 in 2001. Rounded off to closest thousands, the population was _____.

Write the Roman Numerals for the following numbers

- | | |
|--------|--------|
| (a) 6 | (b) 17 |
| (c) 28 | (d) 24 |
| (e) 18 | (f) 22 |

Write the number for the following

- (a) VII (b) IX (c) XI (d) XIV (e) XIX (f) XXVII (g) XXIX (h) XII

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Tick the correct Roman Numerals for numbers:

(a) 24 XIV XXIV XV

(b) 17 VII XII XVII

(c) 14 XIV XVI XV

(d) 29 XIX XX XXIX

Use the symbols > = < in the correct places:

(a) 27 ____ XX

(b) XV ____ IX

(c) 19 ____ XIX

(d) XV ____ 14

(e) V ____ 12

(f) 29 ____ XXX

Perform the operation mentioned and write the answers in Roman Numerals:

(a) $XI \times V =$ _____

(b) $XXV \div V =$ _____

(c) $VII \div II =$ _____

(d) $VII + XXIII =$ _____

(e) $XXX - XXI =$ _____

(f) $XXI - XX =$ _____

Mike forgot the rule while writing Roman Numerals. Help him by drawing a circle around the incorrect Roman Numeral and write the correct numbers beside:

(a) $15 = VVV$

(b) $24 = XXVI$

(c) $8 = IIX$

(d) $19 = XVIII$

(e) $25 = XXVX$

(f) $17 = IIIXX$

Match the Roman Numerals with the numbers:

(a) V 13

(b) XIII 8

(c) VIII 20

(d) XX 5

- (e) XXII 25 (f) XXV 22
(g) XI 11

Detective senses!

Mr Sharma runs the Bulls Eye Bank. You will have to help him, though, because he seems to have forgotten the combination number of the safe of the bank. Fortunately, though, he does remember some information about the number which will help you to help him.

- The number has four digits (eg.2345 or 9376).
- All the digits are distinct (different).
- It starts and ends with an odd number and has two even numbers in between.
- Both 19 and 519 divides the number exactly. Can you find the number which will open the safe?

Match the column

Column 1	Column 2
a. Estimation	p. International System
b. Billion, Million	q. Rough Calculation
c. Crore	r. 90,030
d. Nine Thousand Thirty	s. Indian System

CHAPTER 1

KNOWING OUR NUMBERS

High order thinking questions

1. The population of Agra and Aligarh districts in the year 2001 was 36,20, 436 and 29,92,286, respectively. What was the total population of the two districts in that year?
2. Estimate the product 5981×4428 by rounding off each number to the nearest (i) Tens (ii) Hundreds
3. The town newspaper is published every day. One copy has 12 pages. Every day 11,980 copies are printed. How many pages are printed in total every day?
4. The number of pages available for making notebooks is 75,000. Each notebook contains 200 pages. How many notebooks are possible from the pages available?
5. A machine, on an average, manufactures 2,625 bolts in a day. How many bolts did the machine produce in the month of February 2006?
6. Estimate each of the following calculations by rounding off each number to nearest hundreds:
(a) $874 + 478$ (b) $793 + 397$
(c) $11244 + 3507$ (d) $17677 + 13589$
7. Estimate each of the following calculations by rounding off each number to nearest tens:
(a) $11963 - 936$ (b) $76877 - 7783$
(c) $10732 - 4354$ (d) $78203 - 16407$
8. Estimate each of the products below by rounding off each number to their nearest tens:
(a) 87×32 (b) 311×113
(c) 3239×28 (d) 1385×789

CHAPTER 1

KNOWING OUR NUMBERS

17. The town newspaper is published every day. One copy has 15 pages. Every day 13,560 copies are printed. How many total pages are printed in two days?
18. Rakhi and Sameer both went to the store to buy paper napkins. Rakhi bought 7623 Nitin and Sameer bought 1575 napkins. How many napkins did they buy together?
19. Express each of the following in figures:
 - (i) Twenty-two hundred forty-four
 - (ii) Seven-thousand seven hundred seventy
20. Express each of the following in words:

450	7890
30444	85321

Brain twister

1. If two's company and three's a crowd, what are four and five?
2. If there are five apples and you take away four, how many do you have?
3. Where do aquatic animals keep their money?
4. Two fathers and two sons go fishing. They all individually catch one fish. So why do they have only three fish in total?
5. Mr Watchmen adds five to nine, and gets two. The answer is correct, indeed, but how?
6. The ages of a father and his son add up to 66. The father's age and son's age are reverses of each other. What are their possible ages?
7. What will weigh more - a pound of gold or a pound of bird feathers?

8. If a rooster laid 13 eggs and the poult took eight of them, another rooster laid 12 eggs and four of them were not edible, how many eggs were left?
9. Guess who am I, I am an odd number; take away an alphabet from me, and I become even.
10. Using only sum, how can you add eight 8's to achieve the number 1,000?



Solutions



Basic level

6.	7.	8.
(c)	(c)	(c)

12. 8,000,000



Advanced level

Fill in the blanks

- a. 1
- b. 1
- c. 1000
- d. 10
- e. 10,00,000

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- f. 1000
- g. 1000
- h. 10,00,000
- i. 1
- j. 1650
- k. 1290000m
- l. 422000m
- m. descending
- n. smallest
- o. 999999
- p. 52478401
- q. L
- r. LXVI
- s. 2,538,000

Tick the correct Roman Numerals for numbers:

- a. XXIV b. XVII c. XIV d. XXIX

Match the Roman Numerals with the numbers:

- a. 5 b. 13 c. 8 d. 20 e. 22
f. 25 g. 11

Detective senses!

The combination number of Mr Sharma's bank safe is 9861. A tricky puzzle, indeed, because out of the four clues, the last one alone can give you the answer.

We know that 19 is not a factor of 519, thus, the code for safe has to be a multiple of both 19 and 519.

So simply multiplying 519×19 and you get 9861.

The other clues are just for verifying the answer.

Cross word

a : q

b : p

c : s

d : r

Brain twister

1. 9
2. You took four apples so; obviously, you have four.
3. In the riverbank, obviously.
4. Because the fishing group comprises of a family with grandfather, his son (father), and his son's son - hence there are three people.
5. Add 5 hours to 9 am and you will get 2 pm.
6. There are three possible answers for this: the father-son duo could be 60 and 06 years old, 42 and 24 years old or 42 and 24 years old.
7. Both would weigh the same - after all, a pound is always a pound, regardless of the object!
8. Roosters are males; they don't lay eggs!
9. Seven (SEVEN-S=EVEN)
10. $888 + 88 + 8 + 8 + 8 = 1,000$

END